

ARJUN S. TOMAR

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EDUCATION

PCET's Pimpri Chinchwad University

2027

B. Tech, Computer Science & Engineering (CGPA: 9.1/10)

Pune, Maharashtra

Specialisation in Artificial Intelligence and Machine Learning

- **Relevant Coursework:** Data Structure & Algorithms, Object-Oriented Programming (C++), Operating Systems, DBMS, Deep Learning, Software Engineering Principles

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL

Frameworks: TensorFlow, Scikit-Learn, Keras, PyTorch, LangChain, Hugging Face, YOLOv8

Backend & Tools: Django (MVC), Flask, Celery, Redis, PostgreSQL, Git/Github, Linux (RedHat), ChromaDB

Concepts: Machine Learning, Artificial Intelligence, Data Structures, Algorithms, Computer Vision (CNNs), Transformers, MVC Architecture, Agentic Workflows, Retrieval-Augmented Generation (RAG), Asynchronous Task Queues,

Certifications: RH104 - Introduction to Linux (RedHat), AI Engineering (IBM), Django Web Framework (Meta), Retrieval Augmented Generation (DeepLearning.AI), Vector Databases (IBM), Model Context Protocol (Hugging Face)

ENGINEERING PROJECTS

HireMailer - Async Cold Email Automation Platform

2026

Django, Celery, Redis

- Built a scalable email engine using **Django (MVC pattern)** for the core logic and **Celery with Redis (Memurai)** to handle background task processing, preventing UI blocking during bulk operations.
- Engineered an intentional throttling mechanism (delays) within the task queue to mimic human behavior, ensuring high deliverability rates and preventing IP blacklisting.
- Processed test batches of 100+ emails with status tracking (Pending/Sent/Failed), implementing error handling for SMTP timeouts.

FloatChat - Local Multi-Agent RAG System (SIH 2025 - Team Lead)

2025

Python, Llama 3.2, LangChain, MCP Won the internal college-level round for the Smart India Hackathon (SIH) 2025.

- Successfully orchestrated an agentic workflow running **3 concurrent LLMs locally on 4GB VRAM**, utilizing quantization and model offloading to maintain performance on consumer hardware.
- Implemented a router-based architecture using the **Model Context Protocol (MCP)** to standardize communication between the SQL Agent (query execution) and the Explanation Agent.
- Integrated ChromaDB for efficient context retrieval, reducing hallucination rates by grounding responses in specific dataset schemas.

Handwriting Generation using Sequence Modelling

2025

Deep Learning, LSTM, TensorFlow

- Developed a Long Short-Term Memory (LSTM) network to generate variable-length handwriting strokes from text inputs.
- Built a robust preprocessing pipeline for the IAM-OnDB dataset to handle variable stroke lengths.
- Overcame training instability (vanishing gradients) over 12+ hours of training to produce coherent short-sentence outputs with realistic pen-stroke dynamics.

Autonomous Vehicle Road Sign Detection

2024

- Fine-tuned the **YOLOv8** object detection model on a custom road sign dataset, optimizing hyperparameters to identify traffic signs at simulated high speeds
- Achieved a mean Average Precision (mAP) and confidence score of **>80%** on validation data, balancing inference speed and accuracy for real-time applicability.